Enoch Mok

GitHub | Website | LinkedIn | Email | +65 97343714

Machine Learning Engineer and Data Scientist with experience developing machine learning models, scalable data pipelines, and GPU-accelerated computing infrastructure for healthcare and biomedical research. Proficient in Python, PyTorch, scikit-learn, SQL, Docker, and Linux, with expertise in medical imaging and large-scale data analytics.

Work Experience

Saw Swee Hock School of Public Health, NUS | *Python, Machine Learning, IT Procurement* Mar 2026 – Ongoing
Research Assistant

- Developed and evaluated Python (**PyTorch, scikit-learn, pandas, NumPy**) machine learning pipelines for dementia risk prediction using large-scale demographic population data, achieving an AUC of 0.86.
- Investigating a multimodal deep learning pipeline by integrating blood biomarkers, DICOM MRI imaging, and demographic data, improving the clinical richness of predictive models and supporting ongoing research into multimodal dementia risk prediction.
- Served as the team's Data Manager, developing standardized data dictionaries and managing cross-institutional data requests to harmonize heterogeneous clinical datasets for collaborative research.
- Led the technical specification, procurement, and deployment of $125K in Linux high-performance computing (HPC) infrastructure with **Docker, CUDA, Slurm, SSH**, reducing turnaround time of large-scale workloads by 30%.
- Planned and executed migration of research datasets and computational workloads onto new HPC infrastructure, minimizing downtime and ensuring continuity of machine learning research.

SingHealth | *Python, Tableau, Scripting* Dec 2024 – Mar 2026
Data Analyst

- Built automated **Python** data pipelines and analytics dashboards (**Tableau**, **pandas**, **SQL**) to extract and visualize patient-level metrics for 1.5M+ healthcare records, supporting data-driven decisions and cost optimization.
- Implemented automated data validation frameworks to ensure accuracy and reliability across end-to-end ETL and data pipeline processes, strengthening data integrity for business-critical reporting.
- Partnered with cross-functional teams to optimize data workflows and analytics efficiency, driving $18M+ in cost savings through improved visibility into healthcare performance metrics.
- Led data quality assurance and UAT for a large-scale historical data migration (24M+ rows), resolving inconsistencies to ensure smooth adoption and operational readiness of new analytical systems.
- Applied unsupervised learning, such as k-means clustering and PCA, on 150K+ health survey responses to uncover elderly well-being segments, demonstrating early ML application in population health analytics.

Iota Medtech | *JavaScript, Python, OpenCV* May 2022 – Jul 2022
Software Engineer Intern

- Designed and developed an end-to-end data-driven desktop application for large-scale lung cancer screening using **ElectronJS**, **JavaScript**, **Python**, **OpenCV**, achieving 95%+ accuracy on 1,000+ medical images.
- Implemented secure data management and authentication workflows to ensure data integrity and compliance, aligning with enterprise-grade security best practices.
- Partnered with clinical domain experts to validate the solution, optimizing workflows to screen 1,000+ patients in under 6 minutes and demonstrating measurable operational efficiency through applied data insights.

Projects

Contributor - MONAI, NVIDIA | *Python, PyTorch* | GitHub Apr 2026 - Ongoing

- Enhanced a file integrity validation feature that verifies downloaded artifacts against expected cryptographic hashes, improving the security and robustness of the download workflow by detecting corrupted files.
- Refactored the training framework by introducing reusable base classes and shared utilities, reducing code duplication while preserving backward compatibility and improving maintainability.

HDB-LTA Public Transport Data Analysis | *Python* | GitHub Jul 2024

- Implemented automated data mining **Python** processes to retrieve real-time public transportation information from multiple RESTful APIs and data sources, enabling accurate analysis of current service availability.
- Analyzed public transport statistics to review areas with inadequate transport facilities using **Python**, designing an equity scoring index and geospatial visualizations to illustrate public transport coverage and underserved residential areas.

Federated Learning for Breast Cancer Classification | *Python, PyTorch, OpenCV* | GitHub Jul 2023 - Dec 2023

- Developed and fine-tuned machine learning models in **PyTorch** for mammography DICOM image analysis with **Pydicom**, utilizing big data processing techniques with **OpenCV** and **GradCAM** for AI explanability.
- Simulated a distributed and privacy-preserving training environment using the **Flower** framework, showcasing expertise in scalable data architectures and secure model deployment and optimization.

Education

Nanyang Technological University Aug 2020 – Jun 2024
Bachelor of Engineering - Computer Science *Under CN Yang Scholars Programme*
Specialization: Data Science & Artificial Intelligence

Skills

Languages: English, Chinese (Mandarin), Chinese (Cantonese)
Programming Languages: Python, Bash, SQL, JavaScript
Databases & Cloud: Docker, PostgreSQL, AWS, PouchDB
Visualization Tools: Tableau
Development Tools: Git, Docker